

Effects of land use on soil properties and hydrological processes at the point, plot, and catchment scale in volcanic soils near Turrialba, Costa Rica

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ABSTRACT

Deforestation and conversion to crop land and pasture in tropical regions has important effects on hydrological processes including increased flooding and reduced drought flows. We investigated: 1) differences in soil characteristics between four tropical land uses (forest, coffee, sugar cane and pasture), and 2) how hydrological processes differ between these four land uses. We measured bulk density, saturated hydraulic conductivity (K_{sat}), soil moisture content, catchment discharge and conducted infiltration, mass balance, and dye tracing tests at four catchments within the Reventazón watershed near Turrialba, Costa Rica at the point scale, the plot scale (1 m³), and the catchment scale (1–6 ha). While most of the soil characteristics between the field sites were similar, bulk density at the forest site (0.7 g cm⁻³) was significantly lower than bulk density at the other field sites (1.0 g cm⁻³). At the forest and coffee sites, K_{sat} (> 32 mm hr⁻¹) was at least twice the K_{sat} (< 13 mm h⁻¹) at the sugar cane and pasture sites. During mass balance experiments at the plot scale, sugar cane and pasture had greater surface runoff and lateral flow coefficients (> 30%) than the coffee and forest (< 16%). Irrigated dye experiments revealed roots and macropores were important flow paths in the forest and coffee sites, while machinery, cattle compaction, and lack of large roots impacted the sugar cane and pasture hydrological processes. At the surface, seasonal soil moisture in the sugar cane and pasture sites was consistently greater than in soils of the forest and coffee sites. At the catchment scale, discharge was minimal (< 10%). The sugar cane site had the greatest catchment scale discharge coefficient (7%), event frequency, depth and rate. Overall, the forest and coffee land use had greater water storage capacity, hydraulic conductivity, and percolation compared to the pasture and sugar cane sites. The interplay of different root distributions, compaction, and different soil moisture dynamics affected infiltration and storage dynamics. These findings have important implications for tropical groundwater recharge and runoff predictions.

1. Introduction

In humid tropical countries, hydrological processes profoundly affect economic livelihoods, energy production, human health and the quality of the environment. In Costa Rica, storm runoff has led to loss of human lives, residences, crops, and public infrastructure (Waylen and Laporte, 1999). Extreme peak flows and sediment transport reduce the capacities of the many small scale hydropower projects that produce a majority of the country's electricity demand (Jansson and Erlingsson, 2000; Sanchez-Azofeifa et al., 2002). Percolation, surface runoff, and subsurface lateral flows direct chemical loading that negatively affects water quality as the population increases (Reynolds-Vargas et al., 1994; Rawlins et al., 1998; Renderos-Duran et al., 2002.) Increased

understanding of the effects of land use on hydrological processes that drive flooding, erosion, and contamination will contribute to better water resource management in the humid tropics. Specific knowledge about the hydraulic conductivity, water retention characteristics and rooting patterns of upland crops in tropical environments is needed both in Costa Rica and within the tropical hydrology community (Bonell, 1993; Giambelluca et al., 1999; Bigelow, 2001; Godsey and Elsenbeer, 2002; Bruijnzeel, 2004; Wohl et al., 2012).

Land use affects soil hydrological characteristics such as preferential flow paths, saturated hydraulic conductivity (K_{sat}), macropore connectivity, root mass and water retention (Himo et al., 1987; Hendricks and Flury, 2001; Hanson et al., 2004; Williamson et al., 2004). Preferential flow paths often determine flow directionality (e.g., lateral vs.

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vertical) in humid forested hillslopes (Mosely, 1979; McGlynn et al., 2002; Tromp-van Meerveld and McDonnell, 2006). The K_{sat} can be used to distinguish the dominant hydrological paths in tropical soils (Elsenbeer, 2001; Ziegler et al., 2004). Non-forested land consistently has lower K_{sat} than forested landscapes (Godsey and Elsenbeer, 2002; Zimmerman et al., 2006, 2010; Germer et al., 2010). Compaction from cattle, tractors, plows and foot paths lowers infiltration rates, reduces percolation, destroys macropores, and increases surface runoff (Mendoza and Steenhuis, 2002; Hanson et al., 2004; Ziegler et al., 2006; Verbist et al., 2007). Integrative field campaigns that examine soil moisture dynamics with depth and quantify field hydrological characteristics in the humid tropics are greatly needed to better understand tropical hydrology (Wohl et al., 2012). The specific objectives of this field study were to 1) quantify differences in the hydrologic soil characteristics with depth of coffee, sugar cane, pasture, and forested land uses in a humid tropical region, and 2) compare the extent to which the unique soil hydrologic characteristics of each of these land uses drives hydrological processes at the point, plot, and catchment scales.

2. Methods and materials

2.1. Study site

The study was conducted on the Tropical Agricultural Research and Higher Education Center (CATIE) farm near Turrialba, Costa Rica at approximately 650 masl (Fig. 1). This farm lies approximately in the middle of the Rio Reventazón basin, which is one of the major drainages running from the central mountain range to the Caribbean Sea (Jansson, 1996). The study area receives 2500–3000 mm of rainfall annually (Chacón and Fernandez, 1985). Two volcanoes, Irazu and Turrialba, influence rainfall and soils in the study area through orographic lifting and volcanic ash deposition. During the dry season (January through April), monthly average rainfall ranges from 86.7–177.5 mm. During the wet season (May through December), average monthly rainfall ranges from 245 to 310 mm rainfall. Temperatures average between 20.6 and 22.7 °C throughout the year, with an average annual absolute maximum temperature of 33 °C (December) and an average annual absolute minimum temperature of 10 °C (January). Potential evapotranspiration (PET) ranges from 85 mm month⁻¹ (July) to 115 (April) mm month⁻¹ with a mean of 95 mm month⁻¹.

2.2. Land use characteristics and management

Experiments and monitoring were performed at four different catchments, with each catchment having only one continuous homogenous land use cover: forest (6.1 ha), sugar cane (4.5 ha), pasture (1.9 ha), and coffee (1.2 ha) (Fig. 1). Each catchment had a predominant eastern aspect and flow direction with similar mean slopes and flow lengths (Table 3). These land use types are representative of the predominant land uses for the upper Reventazón watershed (forest 40%, pasture 21%, coffee 14%, and sugar 12%) (Toohey, 2012). Each catchment had homogenous land use with distinct vegetation characteristics and management practices that was drained by a single intermittent stream channel (Fig. 1). At this site, mature sugar cane was burned then harvested in April every year. Following harvest, it is plowed using a large tractor and tillage discs. The sugar cane catchment was nested within the forested catchment (Fig. 1). The forest site contained a mixture of primary and secondary vegetation consisting of few large trees (> 1 m in diameter at 1.5 m in height) and a sparse understory with almost a completely closed canopy. No management had occurred at the forest site since at least the 1950s. The coffee and pasture catchments were hydrologically isolated catchments. In 2004, the pasture was active with approximately 10 cows ha⁻¹ resulting in signs of compaction at the surface by hoof prints and cattle paths that were still noticeable in 2007. After January 2005, when the pasture was abandoned, razor grass with average height of 1 m quickly overtook the pasture field site. The coffee site was managed as a standard *Erythrina* spp. shade system (≈ 50% shaded) with conventional chemical management. Coffee was harvested by groups of workers three times a year. About one month prior to harvest, shade trees were pruned to a tall stump (1–1.5 m) to allow ripening of the coffee beans. Therefore, this site received full sun for about three months of the year.

2.3. Soil properties

Soil sampling was conducted at each field site in January of 2006. Soil sampling was stratified by slope location: toe slope, mid slope and ridge positions. Sampling locations were randomly selected within these stratifications. Nine soil pits (three per slope position) were hand dug in each field site to a 1 m depth. Soils were classified using the National Resource Conservation Service soil taxonomic Handbook (Soil Survey Division Staff, 1993) based on observed soil horizons, color, structure, pore size and density, and rock content of the profile. Root size and depth was measured using calipers and a ruler. Root density

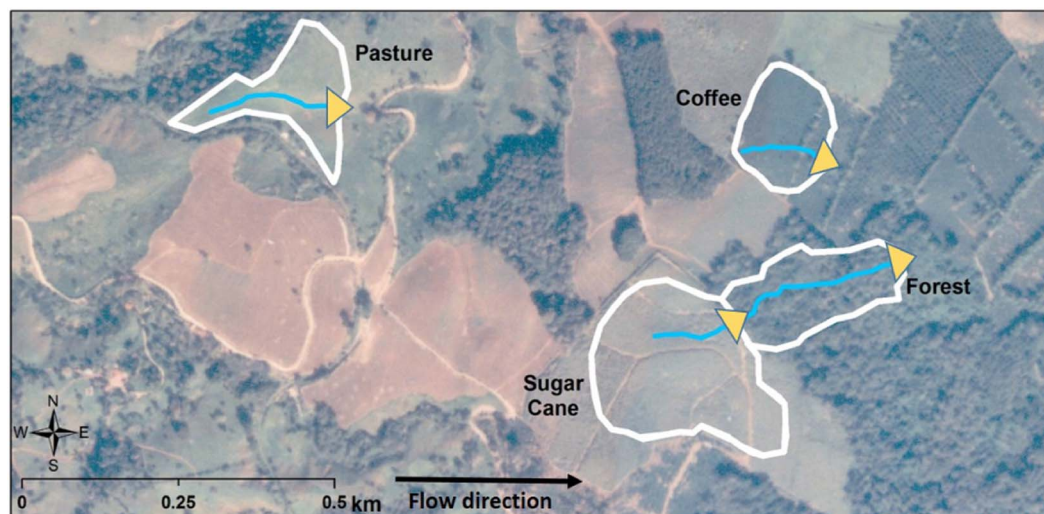


Fig. 1. Location of forest, coffee, sugar cane and pasture catchments (outlined in white). Stream channels are delineated in blue, and flume locations are indicated by yellow triangles. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

was visually estimated along the profile. Soil sampling depths corresponded to soil horizons. Soil samples were taken from the upslope side wall of the soil profiles from 0 to 0.25, 0.25–0.5, and 0.5–1 m depths. Samples were analyzed at CATIE's Soil, Plant Tissue, and Water laboratory for bulk density, organic matter, aluminum and iron oxalate (AlO and FeO) composition (andic properties), and texture. Bulk density samples in duplicate were taken at 0–0.1, 0.40–0.50, and 0.90–1.0 m depths using brass rings with a radius of 31.5 mm and a length of 100 mm. Analysis of variance (ANOVA) was performed to evaluate differences between soil characteristics with depth both within and between field sites ($p < 0.05$).

2.4. K_{sat} and cumulative infiltration

Double ring infiltrometers (diameters of 0.20 m (inner) and 0.50 m (outer)) were used to measure K_{sat} using a steady infiltration rate at a constant head of 35 mm (Reynolds et al., 2002). Similar to soil sampling, the location of these measurements was randomly selected within stratified slope locations (toeslope, midslope and ridge). Thirty measurements at the surface and 21 measurements at the 0.5 m depth were taken in each field site. On flat surfaces ($< 5\%$ slope), vegetation was cut as close to the surface as possible. Leaf litter, twigs, etc., were removed from the sample location. On steeper surfaces and at the 0.5 m depth, careful excavation occurred at the sample location to allow the ring to be level. The steel rings were pushed into the soil to a depth of 50 mm with minimal soil disturbance. Surrounding soil was lightly tamped using the blunt edge of a pencil to avoid flow along the inside wall of the ring. Both rings were filled to 40 mm height with water from a local spring. Once the level of the water within the rings dropped to 30 mm, time was recorded, and the rings were refilled to the 40 mm depth. This process was repeated until the same time was recorded at least three times in a row (2–6 h). At this point, it was assumed that steady-state flow was achieved. The K_{sat} was calculated based on the final time measurement. We used an average of a 35 mm ponded depth in the K_{sat} calculation (Bodhinayake and Cheng Si, 2004).

Saturated hydraulic conductivity was calculated using Eq. (1) (Reynolds et al., 2002):

$$K_{sat} = \frac{\alpha GAQ}{(r(\alpha H + 1 + G\alpha\pi r^2))} \quad (1)$$

where Q is the quasi-steady infiltration rate, A is the area of the inner ring, r is the radius of the ring, H is the height of the ponded water, and α is the Gardner coefficient set at 12 m^{-1} as a general value for agricultural soils (Reynolds et al., 2002). G is a dimensionless geometry factor calculated for $H < 50 \text{ mm}$ (Reynolds and Elrick, 1990):

$$G = 0.316\left(\frac{d}{r}\right) + 0.184 \quad (2)$$

where d equals the depth of the ring insertion (50 mm). Cumulative infiltration depth was calculated as the amount of water required to reach steady state.

2.5. Soil moisture

Soil moisture and temperature was measured using a Davis Complete Soil Moisture/Temperature station (Davis Instruments Corp, CA) installed in each of the field sites above the outlet of the catchment. These stations monitor changes in soil water potential using granular electrical resistive matrix sensors (Watermark Inc., CA). Soil water tension readings were converted to volumetric water content based on an existing soil water release curve for the field site soils (Winowiecki et al., 2007). Two plot locations were selected in each catchment and placed on a similar slope, approximately 2 m from the stream channel and in the mid slope position approximately 25 m upstream from the outlet of the catchment. At each soil moisture site, two sensors were placed at depths of 0.1 m and 1 m. Data records for seasonal soil

moisture cover several months. Data were recorded from 9/17/2006 to 12/15/2006 (forest and coffee), from 11/5/2006 to 1/15/2007 (forest and pasture), and from 12/7/2006 to 2/9/2007 (coffee, sugar cane and pasture).

2.6. Water balance

Three hillslope runoff plots were established along a randomly selected transect at a toe-slope, mid-slope, and ridge location in each field site. Plots were installed in mid-January of 2007 (DOY 18–31). Plots ($\approx 1 \text{ m}^2$) were isolated within the field sites by trenching to a 1 m depth on all four sides. Each soil plot was enclosed in heavy polyethylene plastic on three sides, leaving the down slope side and 0.15 m along the adjacent side wall open. Supports were put into place and trenches were backfilled and compacted. Polyethylene extended 0.1 m above the soil surface to prevent loss of irrigation water. Experiments determined that water retention by the extended polyethylene was consistently $< 0.1\%$ of the irrigation amount. On the open face of each soil plot, metal troughs were inserted at 0.1, 0.25, 0.5 and 1.0 m depths to collect lateral flow from the troughs through a system of funnels and collection containers. Trough depths corresponded to observed soil horizons.

Soil moisture content was measured with WCR probes (WCR_150, Campbell Scientific, Inc., Logan, UT) connected to a data logger (CR 10 $\times$, Campbell Scientific Inc., Logan, UT). The WCR probes were installed into one of the side walls of the runoff plot at 0.1, 0.25, 0.5, and 1 m depths (Fig. 2). All WCR probes were individually calibrated during the field season.

Tarps were placed over each plot during the experiment to minimize effects of natural rainfall and evapotranspiration (ET). In the forest, however, tarps could not cover two of the plots due to the presence of trees. The experiments on the forest plots were conducted on cloudy days with no rainfall during the time of the experiments. Prior to each water balance experiment, plots were irrigated during the afternoon of

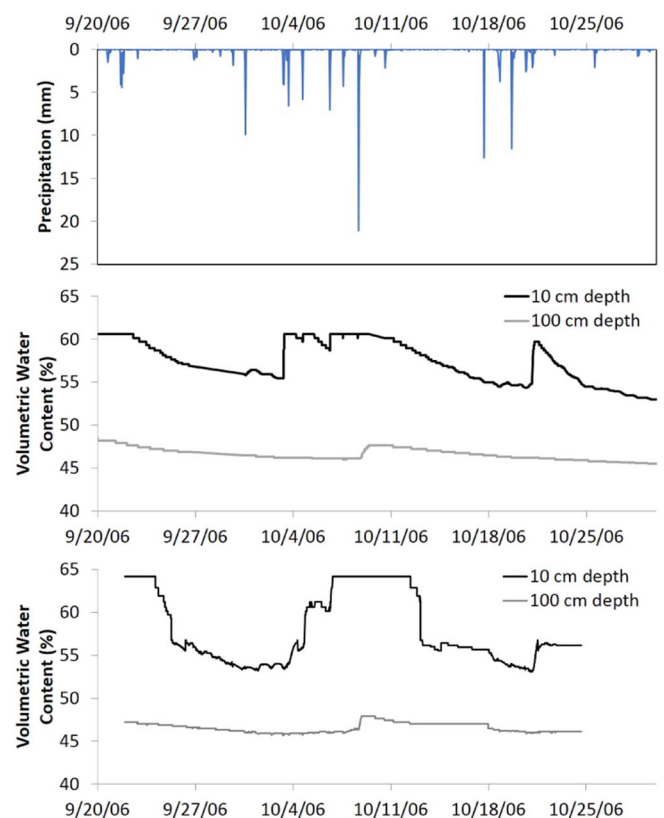


Fig. 2. Precipitation (top), forest (middle) and coffee (bottom) volumetric soil water content.

the previous day for at least 3 h at 20 mm h^{-1} intensity to reduce differences between plots due to ET. Water was added by two backpack sprayers constantly moving for 15 min. Irrigation was applied to all plots at an initial rate of 20 mm h^{-1} for at least 1 h duration. During the second and third hours of the irrigation, rates were incrementally increased to approximately 40 and 60 mm h^{-1} , respectively. If runoff rates were minimal at the 60 mm h^{-1} rate then rainfall rates were increased incrementally up to a point where consistent surface runoff occurred. For the forested land use the maximum application rate required to generate substantial surface runoff was 220 mm h^{-1} (see Table S3).

Within the top 1 m of soil within each plot, a water balance was calculated using the applied irrigation water depth, lateral flow collected in the troughs, changes in soil moisture storage, and the assumption that ET losses were negligible. The water balance equation over the duration of the experiment was solved for percolation below the 1 m depth as follows:

$$D = P - \Delta S - R - L + e \quad (3)$$

where D is percolation (L) P is irrigation water (L), ΔS is soil moisture storage (L), R is surface runoff, L is lateral flow from the soil column, and e is the measurement error. Errors in the water balance terms were estimated as $0.05 \text{ (L}^3\text{L}^{-3}\text{)}$ for volumetric water content (Campbell Scientific, 2004) and 5% for lateral flow (Redding and Devito, 2008).

2.7. Dye experiments

For the final runoff experiment at each plot in each field, 3.0 g of Brilliant Blue Dye was added to and mixed within the backpack sprayers. At the end of the experiment, the plots were excavated from the top down every 0.1 m with a wide machete. Photos were taken, structural characteristics were noted, and depth of dye was recorded. Root diameters, depth from the surface, were also measured in three dimensions. Root density was visually estimated based on NRCS Soil Survey protocols.

2.8. Discharge

Four H-flumes (Brakensiek et al., 1979) were installed in incised intermittent channels at the outlet of each of the catchments to measure catchment-scale discharge between July 2004 and December 2006. Pressure Transducers (Druck, ISCO) and CR10X data loggers (Campbell Scientific, Logan, UT) were installed to record water level in each flume. Due to extreme weather events, vandalism, or instrument malfunction several gaps exist in the data set as follows: July 2004–April 2005 (sugar cane), 1/1/05 to 1/15/05 (all sites), 1/24/06 to 2/05/06 (coffee), 7/3/06 to 7/12/06 (pasture), 7/12/06 to 8/26/06 (sugar cane), 7/31/06 to 8/12/06 (forest), and sporadic readings from 7/3/06 to 8/14/06 (coffee). Hydrograph analysis only occurred when all four catchments had functioning instrumentation.

Hydrographs were analyzed to determine total discharge depth, peak flow, event frequency, peak discharge rates, and discharge coefficients. Discharge coefficients were determined as the ratio of discharge to rainfall for the designated period of time (Dingman, 1994). Rainfall analysis was based on 15-minute tipping bucket rainfall data. Peak discharge was determined from 15-minute discharge data. Event response time was defined as the time between the initiation and cessation of runoff. To perform hydrograph analysis on the forest catchment, sugar cane discharge was subtracted from the forest discharge to isolate discharge produced by the forest catchment.

3. Results

3.1. Rainfall

Annual rainfall from 2004 to 2006 was 2850, 3159, and 2760 mm

Table 1

Rainfall characteristics at the CATIE meteorological station from 2004 to 2006.

Rainfall event volume	Frequency	Event duration	Frequency	Max. 15 min rainfall	Frequency
(mm)		(h)		Intensity (mm h^{-1})	
> 200	5	> 200	1	> 60	4
> 100	9	> 100	12	> 40	33
> 50	34	> 50	44	> 20	222
> 10	178	> 10	267	> 10	658
> 1	2816	> 1	2816	> 1	1620
Maximum	448 mm	Maximum	233 h	Maximum	120 mm h^{-1}

respectively. From 1942 to 2016, mean annual rainfall was 2697 (SD = 500) mm. Therefore, 2005 was a relatively wet year compared to this historical record. Rainfall events were defined as the time between the initiation and cessation of rainfall with a lack of rainfall for at least 15 min. All reported rainfall intensities are maximum 15-minute intensities. The maximum recorded rainfall intensity over the study period was 120 mm h^{-1} (Table 1). The observed historical maximum daily rainfall was 429 mm. The maximum rainfall event volume from 2004 to 2006 was 448 mm. From 2004 to 2006, over 250 rainfall events (10%) experienced rainfall intensities ($> 20 \text{ mm h}^{-1}$) similar to those irrigation intensities used in the plot scale experiments. Only two of the irrigation experiments (at the coffee and forest site) exceeded the historical maximum daily rainfall (Table S3).

3.2. Soil properties

Soil sampling indicated each field site was located on an Inceptisol with three different horizons. A dark A-horizon extended to about 0.2 m. An AB-horizon transition layer was identified from about 0.2 m to 0.6 m, followed by a reddish B-horizon. A drainage ditch excavated on the border of the coffee field site showed the B-horizon extending to at least 2.5 m. An existing embankment nearby the forest field site showed the B-horizon extending to at least 5 m. Bedrock was observed at about 5 m depth in an old railroad cut just east of the pasture site.

Clay content, organic matter and andic properties were similar at all field sites (Table 2). At the surface, clay content was $< 31\%$ and significantly increased to at least 44% with depth ($p < 0.05$). Organic matter was $> 5\%$ at the surface at each field site and decreased with depth to approximately 2% at 1 m. At the surface, AIO and 1/2 FeO in each of the field site soils was either just above or slightly below the threshold of 1.0% to be classified as having andic properties (Soil

Table 2

Soil characteristics at different depths for each field site.

Field site	Depth	Bulk density	Porosity	AIO + 1/2FeO	Soil organic matter	Sand	Silt	Clay
	(m)	(g cm^{-3})	(%)	(%)	(%)	(%)	(%)	(%)
Forest	0.05	0.73a	72	0.90a	6.5a	57	21	22a
Forest	0.5	0.81a	69	0.94a	4.3a	38	22	40a
Forest	1.0	0.95b	64	0.86a	2.5b	32	18	50b
Coffee	0.05	0.92b	65	0.89a	5.3a	48	22	30a
Coffee	0.5	0.86b	68	0.7b	3.4a	33	19	48b
Coffee	1.0	0.94b	65	0.62b	1.6b	30	17	53b
Sugar	0.05	0.99bd	63	1.2a	5.5a	51	24	24a
Sugar	0.5	0.91bd	66	1.0a	4.5a	42	23	35a
Sugar	1.0	0.99bd	63	0.89a	1.8b	38	18	44b
Pasture	0.05	0.87bc	67	1.0a	5.3a	47	28	25a
Pasture	0.5	0.87bc	67	0.70b	4.1a	34	30	36a
Pasture	1.0	1.00bd	62	0.60b	2.2b	27	24	48b

For each soil characteristic, different letters denote a statistically significant difference.

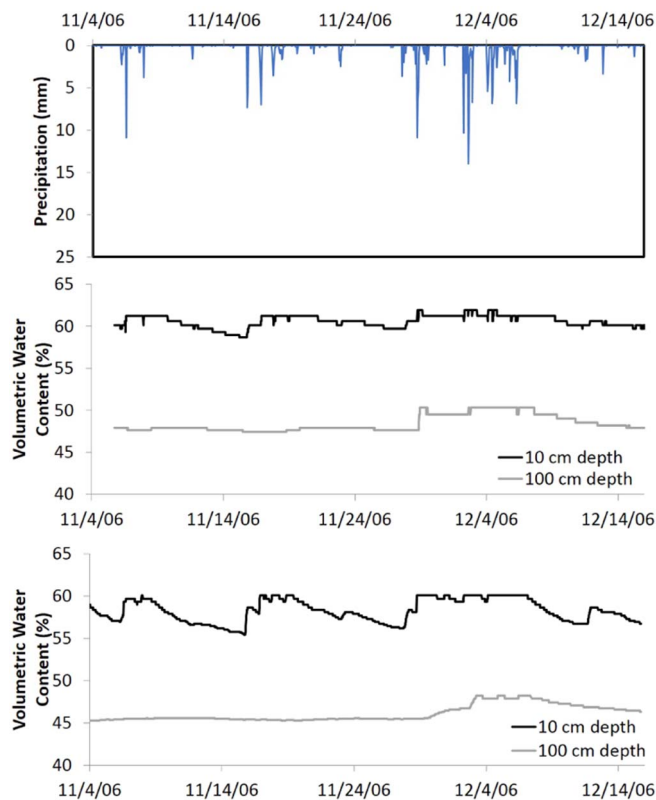


Fig. 3. Precipitation (top), and pasture (middle) and forest (bottom) volumetric soil moisture.

Survey Division Staff, 1993). Percentage of rocks was extremely low at all field sites (0–5%). The bulk density of the forest field site at the surface (0.73 g cm^{-3}) was significantly lower ($p < 0.05$) than surface bulk density at the other three field sites (Table 2). Also, surface bulk density at the pasture site was lower than surface bulk density at the sugar cane site. For both pasture and forest, bulk density was significantly greater at 1 m depth than at the surface. With the exception of the differences in bulk density at the forest and pasture site, the matrix soil was very similar between all the field sites at the point scale of measurement.

3.3. Seasonal soil moisture

During the period of soil moisture monitoring (9/2006–2/2007), soils in the pasture field site were consistently wetter than soils in the other land uses (Figs. 3–4). When rainfall was minimal, moisture contents were lowest at the forest and coffee sites (Figs. 2–4). At the 0.1 m depths, the soil moisture recession curves were steeper in the forest and the coffee sites than in the pasture and sugar cane sites suggesting differences in percolation and/or evapotranspiration (Figs. 2–4). At all land use field sites, the 0.1 m soil moisture increased rapidly when rainfall occurred. At the 1.0 m depth, much smaller variations in soil moisture content occurred at all field sites. However, within the coffee and forest sites, soil moisture increases at 1.0 m occurred anywhere from 24 to 96 h after similar increases occurred at the sugar and pasture sites suggesting slower infiltration or greater soil moisture deficits within the coffee and forest sites.

3.4. K_{sat} and cumulative infiltration

The surface K_{sat} in the forest and coffee field sites were over two times greater than K_{sat} in the pasture and sugar cane sites (Fig. 5, Table S1). The forest and coffee field sites also exhibited the greatest amount

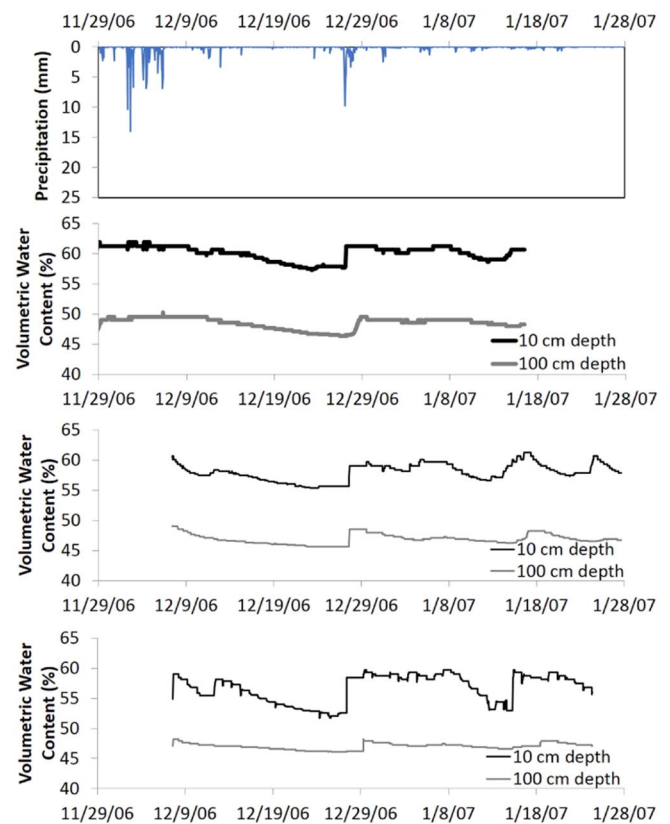


Fig. 4. Precipitation (top), and pasture (second panel), sugar (third panel) and coffee (bottom) volumetric soil water content.

of variance in K_{sat} at the surface with several outliers reaching above 300 mm h^{-1} (Fig. 5). At 0.5 m, K_{sat} mean values in the forest, coffee and pasture field sites were also greater than in the sugar cane sites. The K_{sat} values measured at 0.5 m were less than K_{sat} values measured at the surface, except for the pasture K_{sat} at 0.5 m which was greater than K_{sat} at the surface (Table S1). Sugar cane and pasture had lower variance of K_{sat} when compared to the forest and coffee.

Similar to the results from the continuous soil moisture monitoring, measurements of the cumulative infiltration to reach K_{sat} indicated that the woody vegetation land uses were considerably drier ($> 19 \text{ mm}$) than the herbaceous sugar cane and pasture sites (Table S2). Despite lower K_{sat} in coffee, cumulative infiltration in coffee at the surface was greater than in forest. These greater values may be the result of increased soil evaporation at the coffee site similar to the observations provided by the continuous soil moisture sensors.

3.5. Water balances

Plots in the sugar cane and pasture field sites had much greater surface runoff and lateral flow as a percentage of applied water than plots in forest and coffee field sites (Fig. 6, Table S3). Sugar cane surface runoff, which mostly started within the first hour of irrigation, was $> 25\%$ of the applied irrigation with varying amounts of lateral flow from subsurface layers (Table S3). Most of this surface runoff occurred on the compacted rows between the sugar cane, or approximately 50% of the plots. Frequently, lateral flow at 0.25 m depth constituted about 5% of the applied irrigation water in the sugar cane, but arrived later than the surface runoff. Soil moisture at the 0.5 m and 1.0 m depth responded relatively quickly after the initiation of irrigation indicating that water was delivered to the deeper soil matrix.

In the pasture field site, lateral flow preceded surface runoff. However, surface runoff eventually increased to be a greater proportion (14–35%) of the applied water suggesting a limited conductivity in the

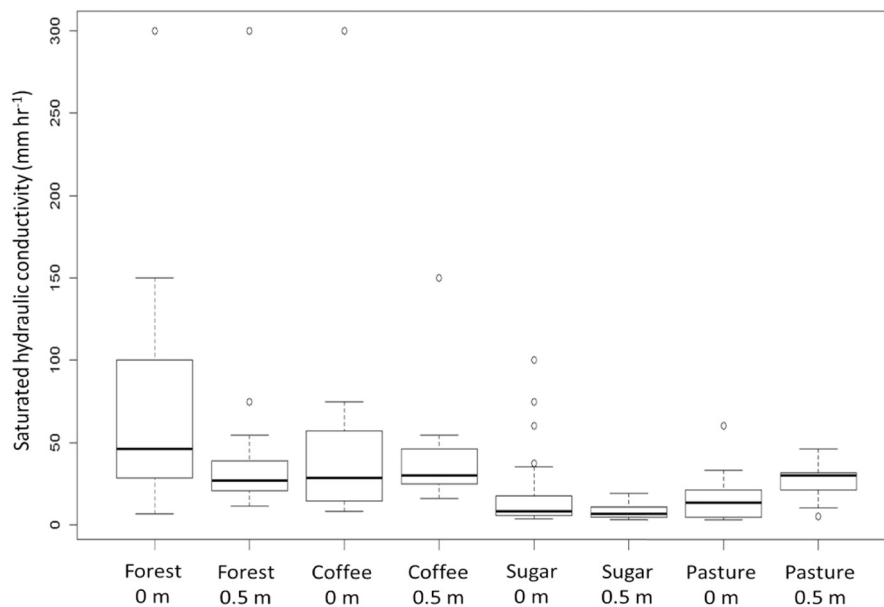


Fig. 5. Box and whiskers plots of saturated hydraulic conductivity for each field site at the surface and 0.5 m depth (open circles represent 1.25 quantile outliers and closed circles represent 1.75 quantile outliers).

deeper soil horizons. In the pasture, lateral flows down to 0.5 m accounted for 14–32% of the applied water (Fig. 6 Table S3). During these experiments surface runoff appeared to result from a perched water table between 0.25 and 0.5 m in the pasture. At 0.5 m, soil moisture increased rapidly even during the lowest irrigation intensities. Lateral flow at this depth began when soil moisture approached saturation. As irrigation intensity increased, surface runoff, and lateral flow increased. In addition, surface runoff was limited by the increased surface roughness from hoof compaction and high vegetation density of the razor grass. Both of these factors limited hydrological connectivity on the surface within the pasture sites. The combination of the surface runoff, lateral flows, and change in storage during the experiments severely limited percolation in all three of the pasture plots (< 15%) (Fig. 6). During wet periods with high intensity rainfall, percolation in the pasture field site may be limited as a larger portion of the water leaves the plot as surface runoff and lateral flow due to impeding hydraulically restrictive soil horizons.

While the coffee and forest plots experienced some surface runoff and lateral flow, these amounts were relatively small (< 5% surface runoff and < 15% lateral flow) and only occurred after longer duration irrigation (4–6 h). Lateral flow generation began before surface runoff in all coffee and forest plots (0.25–0.5 m depth) suggesting a saturation excess runoff mechanism. Both increasing storage and vertical percolation were the dominant hydrological process at the coffee and forest plots.

3.6. Dye experiments

Root size, density, and vertical distribution differed among the four land uses. At the coffee and forest field sites, roots were much larger (up to 50 mm in diameter) than in the pasture and sugar cane sites (Fig. 7). At the pasture and sugar field sites, many small roots were present in the upper 0.1 m of the soil. The pasture field site had the smallest roots (1 mm diameter) followed by sugar cane roots (5 mm diameter). A

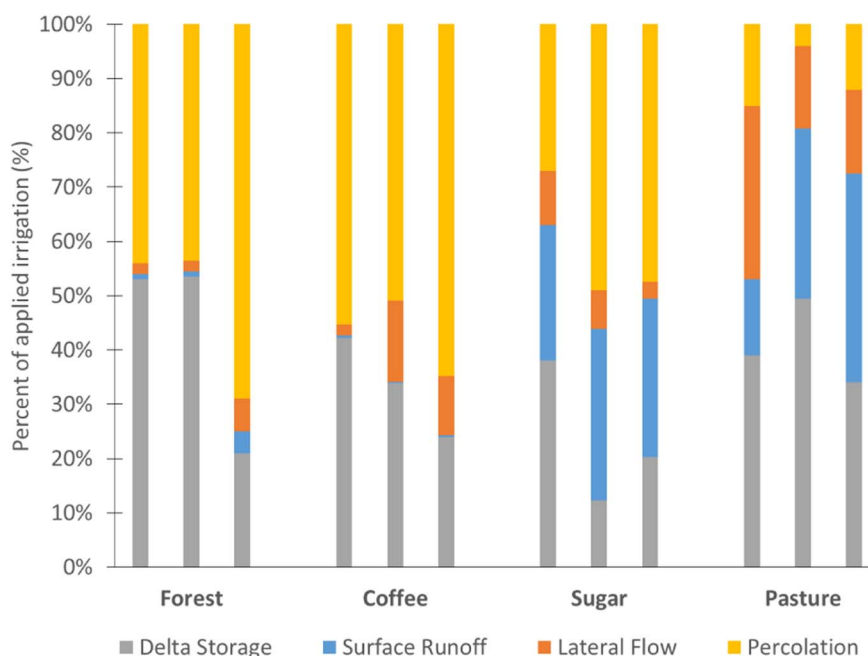


Fig. 6. Results from the water balance experiments indicating that storage and percolation were dominant hydrological processes in the woody land uses whereas surface runoff and lateral flow were dominant hydrological processes in the sugar cane and pasture sites.



Fig. 7. a) Coffee plot dye experiment (each line on ruler equals 0.1 m) b) Pasture plot after the dye experiment (lines on the ruler equals 0.1 m). c and d) Sugar plot after the dye experiment demonstrating evidence of a plow layer.

0.1–0.15 m thick vegetative mass in the pasture field site extended above the soil surface. Root density decreased with depth at all four sites. Most commonly, only a few fine roots (1 mm) were found at a depth of 0.5 m and greater in the sugar cane and pasture site. However, at the coffee and forest sites, large roots (5–50 mm) were found throughout the 1 m profile. The forest had the largest roots and the greatest root density. Root size and distribution were distinctly different between the herbaceous and woody vegetation land uses (Fig. 7).

Dye stained roots in the forest and coffee plots carried water to greater depths than in the sugar cane and pasture plots (Fig. 7). In Fig. 7b, even redistribution of dye occurs within the top 10 cm of the pasture, whereas in Fig. 7a, the coffee has a much greater depth of redistribution approaching 30 cm depth. While the majority of the dye was found within the upper 30 cm depth in all land uses, soil moisture at 0.5 and 1.0 m sensors also increased. In forest and coffee plots, dye stained the large, thick roots and their surrounding soils concentrated in the upper 0.5 m and several roots preferentially down to depths of 1.0 m (Fig. 7b). Dye stains in the sugar cane and pasture plots showed a

more uniform pattern of small roots up to approximately 0.5 m, with very little dye moving past this depth (Figs. 7a). Dye experiments showed roots and other preferential pathways as evidence of both lateral and vertical water transport at the coffee and forest field sites. These roots and macropore networks in the woody vegetation land use sites represent a major difference from the herbaceous land use sites.

Signs of compaction within the pasture and sugar cane site were also visible as the result of the dye experiments (Fig. 7b, c, and d). As previously mentioned within the pasture sites, the majority of dye did not infiltrate further than 10 cm depth (although smaller preferential pathways carried the dye deeper). Within the sugar cane, specifically within the rows between the plants, surface compaction appeared to limit infiltration of dye to approximately 10 cm depth. Additional evidence of subsurface compaction due to a plow layer was observed between 25 and 35 cm depth within the sugar cane plots (Fig. 7c and d).

Table 3

Catchment morphological and discharge characteristics (for 5/2005 to 6/2006 and 9/2006 to 12/2006).

Catchment morphological characteristics	Forest ^a	Coffee	Sugar cane	Pasture
Catchment area (ha)	6.1	1.2	4.5	1.9
Mean flow length (m)	1778	609	1133	773
Mean catchment slope (°)	8	7	4	11
1.0				
Catchment discharge characteristics	Forest	Coffee	Sugar cane	Pasture
Rainfall (mm)	3826	3826	3826	3826
Total discharge depth (mm)	79	114	262	49
Event frequency (n)	100	134	195	37
Peak discharge rate (mm hr ⁻¹)	11.1	1.7	34.7	11.7
Peak discharge depth (mm event ⁻¹)	37.4	6.5	51.8	12.2
Discharge coefficients	2%	3%	7%	1%

^a Sugar cane catchment was nested within the forested catchment.

3.7. Stream discharge

Stream discharge at the catchment scale differed between land uses with respect to event frequency, peak discharge rate, peak discharge depth, and discharge coefficient (Table 3). The sugar cane responded rapidly to high rainfall intensities and/or amounts (Figs. S1 and S2). Overall, the sugar cane field site had the greatest frequency of discharge events, peak discharge rate, peak discharge depth and discharge coefficient (Table 3). Interestingly, the pasture catchment had the lowest number of events, total discharge depth and discharge coefficient when compared to the other catchments. However, it had the second highest peak discharge rate (11.7 mm h⁻¹) after sugar cane. The pasture site rapidly responded to high rainfall intensity and/or additional rainfall during wet periods of the year after surface detention was exceeded (Figs. S1, S2). Pasture discharge response curves had high discharge rates and short durations. When rainfall intensity decreased pasture discharge quickly subsided (Fig. S1). However, compared to the other land uses, discharge occurred less frequently at the pasture site which can be partially attributed to the surface water detention process (Table 3, Fig. S2). Within the coffee catchment, peak discharge rate and peak discharge depth are both much less than at the other field sites. However, coffee had the second highest total discharge depth, number of events, and discharge coefficient. Similar to the other infiltration data, the infiltration capacity of the forest site was very large at the catchment scale. The forest field site had the second lowest discharge depth, event frequency and peak discharge rate (Table 3). However, the forest experienced the second largest peak discharge depth (37.4 mm). When leaving the forest discharge record unaltered, approximately 63% of the sugar cane discharge (12,100 m³) re-infiltrated in the forest (4449 m³) (Fig S3). Several large rainfall events occurred, however, where discharge measured below the forest site was greater than below the sugar cane site (Fig. S3). We hypothesize that as the variable source area drained within the sugar cane, discharge velocity and amount decreased enabling re-infiltration into the forest field site.

4. Discussion

4.1. Semi-restrictive soil layer

While distinct differences in hydrological properties and soil moisture dynamics were observed between the different land uses, the soils at each field also had characteristics that supported the development of a hydrologically semi-restrictive layer around 0.5 m. The soils at each field site represented a typical profile for an older, tropical, deep soil (~5 m) with little rock content. Clay content increased to approximately 50% at 1 m depth. Also, high organic matter contents

(> 5%) and andic properties decreased with depth. Andic properties and organic matter content contributed to low bulk densities (< 1.0 g cm⁻³) even though clay contents were high. Andic properties greatly increase the surface area of soils due to the size and structure of volcanic ash and glass. These substances can combine with organic material in soils to form structurally stable volcanic ash aggregates (Shoji et al., 1993). Andic properties explain why porosity and infiltration capacities are substantially greater in these soils than other tropical clayey soils (Dahlgren et al., 2004; Jiménez et al. 2006). During the runoff experiments, subsurface lateral flow was initiated between 0.25 and 0.5 m at each plot. The decrease of Andic properties, organic matter and roots along with the increase of clay content, most likely formed a semi-restrictive layer at depth in all land uses due to reduced porosity, preferential pathways and ultimately lower K_{sat} values. The similarities of the soil matrix properties at the different field sites provided the opportunity to determine what impact the individual land uses had on hydrological processes within deep tropical soils that have the unique ability to infiltrate and store large amounts of water.

4.2. The role of soil compaction

The lower bulk density in the forest was the only significant difference between soil matrix properties at the various field sites. In similar studies, forested sites often have much greater organic matter content than pasture that generally leads to lower bulk densities and greater infiltration capacities (Solomon et al. 2000; Richter and Markewitz, 2001; Bonell et al., 2010; Don et al., 2011). Spaans et al. (1989) found similar results in Costa Rican soils north of our field sites when comparing bulk densities before and after forest clearing. This lower bulk density in the forest suggests evidence of compaction in the sugar cane, coffee, and pasture field sites. Each of these sites received foot, cattle, and/or machinery traffic. Many studies have shown that forest soils have lower bulk density than soils in other land uses due to less human and cattle induced compaction (Lee and Foster, 1991; Ziegler et al., 2006; Price et al., 2010). The andic nature of these soils also suggests they may be particularly susceptible to compaction when physical aggregate degradation occurs. The lower bulk density of pasture soils is most likely due to the abandoned nature of the pasture and the dense, fine root network of the pasture near the surface of the profile that developed when razor grass overtook pasture grasses. This mat had very high porosity and the ability to retain surface water. At the same time, shallow compaction from hoof prints also contributed to large amounts of surface water detention and roughness similar to Russell et al., 2001. Both vegetation density of the razor grass and increased surface roughness likely increased surface water storage and limited overland flow connectivity in the pasture catchment.

In terms of infiltration, the sugar cane had the lowest K_{sat} at the surface and at 0.5 m depth. The sugar cane field site was influenced by burning, tillage, and compaction. During the plot experiments, the sugar cane produced the most surface runoff (~30%). Although lateral flow occurred at the sugar cane plots, it occurred later than surface runoff and the volume was much less (< 10% of irrigation). Interestingly, lateral flow most often was generated at the 0.25 m depth before deeper horizons. From the dye application, a plow layer was observed at this depth which likely explains the lateral flow generation. Jiménez et al. (2006) determined that anthropogenic disturbance (e.g., burning, tillage and air drying) significantly lowered the naturally high infiltration capacities of andic soils and increased surface runoff. The tillage discs, used at the sugar cane site, might cause subsurface compaction and the formation of a plow layer at the 0.25 m depth. The destruction of andic aggregates and macropore structure through crop management practices of sugar cane could contribute to the lower K_{sat} and greater surface runoff in the sugar cane. In addition, the sugar cane plots appeared to exhibit infiltration-excess runoff at higher rainfall intensities (< 20 mm h⁻¹).

At the pasture site, infiltration, surface runoff, and lateral flow

generation seem to be influenced by soil compaction and the lack of large roots and macropores, organic matter and andic properties with increasing depth. The K_{sat} at 0.5 m depth was two-fold greater than at the surface in the pasture site. At the plot scale, the pasture site clearly indicated a saturation-excess runoff generation as lateral flow initiated at 0.5 m similar to a Variable Source Area process driven by differences in hydraulic conductivity (Germer et al., 2010). Irrigation amount and intensity needed to exceed soil storage and conductivities to generate runoff in the pasture and sugar. Therefore, observed discharge events at the pasture field site may be reflecting continued rainfall on top of a saturated soil column. In addition, the surface roughness due to cattle compaction may have greatly increased surface water detention within the pasture. This may also explain why bulk density in the pasture did not indicate signs of compaction while K_{sat} was negatively impacted.

4.3. The role of root distribution

Hydrological processes such as infiltration, percolation, runoff generation, and lateral flow differed greatly between the land uses at the point and plot scales. Our observations suggest that these differences were due to root, macropore and preferential pathway size and distribution. The forest and the coffee had the greatest K_{sat} , the greatest infiltration rate, and the least amount of surface runoff and lateral flows at the point and plot scales. Upon excavation of the forest and coffee plots after dye application, the dye reached greater depths than in the sugar cane or pasture plots by using thicker roots and/or preferential pathways down to at least a 1 m soil depth. The forest and coffee plots had much larger roots at greater depths than the sugar cane and pasture. In the forest and coffee, as surface soil storage became saturated, vertical macropore networks were activated that promoted percolation to increase subsurface connectivity. Tree roots can promote wider and deeper preferential flow paths than cultivated cropland or pasture (Martinez-Meza and Whitford, 1996; Yunusa et al., 2002). Roots create preferential pathways through the combination of localized compaction by root growth and the addition of root exudates to the adjacent soils (Johnson and Lehman, 2006). Root death and decay is another plausible cause of preferential flow paths. In each land use, root density and preferential flow paths decreased with depth from the surface. Root zones can increase K_{sat} relative to the soil below the root zone. This gradient of conductivities often determines flow path direction in terms of percolation, lateral flow, or saturation-excess overland flow (Beer et al., 1997; Elsenbeer, 2001). The decrease of organic matter and andic properties with depth may further drive this gradient by limiting aggregate formation and thus storage capacity and preferential pathway formation into deeper soil horizons. At the coffee and forest sites, greater amounts of water percolated downward than at the sugar cane and pasture sites during the runoff experiments. Gomez-Delgado et al. (2011) found similar percolation amounts through modeling simulations in a nearby coffee basin. The lack of these preferential pathways in the sugar cane and pasture likely contributed to the generation of surface runoff and lateral flow during the runoff experiments.

Several of the K_{sat} values for the forest and coffee field sites were $> 300 \text{ mm h}^{-1}$ (Fig. 5). While these values were treated as outliers in our analysis, these outlier values likely were caused by preferential pathways located under the ring infiltrometers. Several K_{sat} outlier values were also observed in the sugar cane site (Fig. 5). These coincided with measurements taken over the actual sugar cane plants (not on the rows between) suggesting a bypass flow mechanism when compared to the compacted rows in the sugar cane similar to Sobieraj et al. (2004). Variance of surface K_{sat} values in the forest and coffee indicates that the greatest macroporosity occurs in these woody vegetation land uses. At each of the field sites, variance in K_{sat} decreased with depth suggesting that the relative effect of land use on K_{sat} declined below the root zone possibly due to limited root access and decreased macroporosity.

4.4. Upscaling to catchment discharge

The apparent differences in hydrological flowpaths between the field sites at the point and plot scales diminished at the catchment scale where runoff coefficients were similar (1–7%). The magnitude of discharge was similar to runoff coefficients measured in coffee near our study sites (Gomez-Delgado et al., 2011). Even in tropical humid climates, studies show that annual runoff coefficients are frequently $< 10\%$ of rainfall in small catchments (Jansson, 2002; Cattani et al., 2006). However, these small discharge coefficients still translate to over 260 mm of annual discharge. In general, much of the literature reports that deforestation and conversion to agricultural land uses increases river discharge and groundwater levels due to the loss of the evapotranspiration of the forest (Bosch and Hewlett, 1982; Farley et al., 2005). However, some studies in humid regions have shown similar ET rates between woody and herbaceous land uses (Brauman et al., 2012; Kelliher et al., 1993). While we did not directly measure evapotranspiration in each catchment, PET accounted for $> 50\%$ of rainfall from 2004 to 2006. Also, woody-based land uses often have greater interception rates than their agricultural counterparts (Macinnis-Ng et al., 2012; Muñoz-Villers and McDonnell, 2013; Safeeq and Fares, 2014). Arguably, we observed differences in soil moisture between the catchments that could be due to interception, evapotranspiration and/or percolation. However, seasonal soil moisture in each catchment responded similarly to rainfall events by wetting up with similar rates and amounts suggesting small differences in interception between the land uses. The woody-based land uses often had relatively drier surface soils between rainfall events suggesting greater differences in ET and/or percolation than the herbaceous land uses. However, deeper soil moisture within the woody-based land uses increased relatively later (up to 96 h) than the herbaceous land uses suggesting greater soil moisture deficits at the surface of the woody-based land uses. Similar to Hanson et al. (2004), percolation at the pasture field site was limited by lack of macropores. Storage capacity of these andic soils and vertical flow paths still appear to be dominant parts of the hydrological cycle similar to Salemi et al. (2013). During the same irrigation period, the forest and coffee field sites were able to percolate more water suggesting that under large storm events, these woody-based land uses may indeed have the ability to recharge groundwater supplies. Soil storage, percolation and groundwater dynamics must play important roles as runoff coefficients of larger watersheds in Costa Rica are upwards of 60% (Toohey 2012). The subsurface hydrological connection between intermittent catchments and larger perennial streams and rivers needs further investigation.

5. Conclusions

Lower bulk density at the surface of the forest was the only observed soil matrix difference due to land use in this study. However, all land uses had relatively low bulk densities due to high organic matter ($> 5\%$) and the presence of andic properties in this soil to a depth of 0.5 m. Therefore, in the top 0.5 m of these soils, soil water storage capacity was very high. Below 0.5 m, clay content significantly increased while organic matter and andic properties decreased. The difference between these two layers provided for a semi-restrictive layer within the AB horizon ($\approx 0.5 \text{ m}$).

At the surface, soil moisture in the woody land uses (i.e., forest and coffee) was often less than the herbaceous land uses (i.e., pasture and sugar cane) due to differences in drainage and/or evapotranspiration. When rainfall was minimal, soil moisture drying curves were steeper and more sustained relative to the grass based land uses. The often exposed soil surface (e.g., after herbicide application) of the coffee field site was the driest land use with the steepest soil moisture drying curves suggesting soil evaporation also played a role in these differences. All of the land uses' moisture contents rose quickly in response to rainfall. Differences in soil moisture between land uses at 1.0 m depth was much

less than at the surface. However, increases in soil moisture at the 1.0 m depth were relatively slower in the woody-based land uses with greater soil moisture deficits.

Point and plot scale measurements showed that infiltration, lateral flow, surface runoff and percolation were strongly influenced by land use. Compaction and/or burning and tillage limited infiltration at the sugar cane site to the point of causing infiltration-excess runoff at increased rainfall intensities ($> 20 \text{ mm h}^{-1}$). At lower precipitation intensities, we speculate that subsurface compaction at the plow layer contributed to runoff generation through lateral flow. The woody based land uses had much larger preferential pathways and roots that appeared to greatly increase infiltration and percolation at the point and plot scales.

At the catchment scale, land use affected runoff during the wet season. During the dry season, runoff events were extremely limited. During the wet season, sugar cane had the greatest observed runoff event depth, frequencies and rates. The majority of this response was most likely due to infiltration-excess runoff. Pasture events were rare due to surface water storage created by hoof compaction and dense herbaceous vegetation that overtook the pasture site when it was abandoned. When pasture runoff events did occur, they generally occurred in response to high intensity and short duration rainfall due to a saturation excess mechanism moderated by the storage capacity of the vegetative mat and cattle induced surface roughness at the surface. Based on the plot experiments in the pasture, percolation was limited during extreme rainfall events with most water being diverted laterally as subsurface flow and runoff. Limited percolation during these events may limit groundwater recharge. During the wettest periods, coffee and forest both appeared to deliver runoff when precipitation volume was high. The majority of coffee and forest runoff events had lower discharge rates than the sugar cane or pasture. During extreme irrigation events in these woody land uses, most water was directed vertically as percolation in our plot experiments. Due to nested nature of the sugar cane within the forested catchment, we observed that the forest site had an immense ability to re-infiltrate sugar cane runoff. Therefore, woody-based land uses positioned below runoff prone land uses may contribute to some mitigation of flooding and recharge of aquifers. Finally, lateral flow or return flow, via lateral conductivity due to roots and macropores, may be a subtle but important runoff mechanism for woody based land uses during very wet periods. More replication and detailed study at the landscape scale due to the heterogeneity of these systems should be conducted to properly assess this potential.

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Appendix A. Supplementary data

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